

MH2500 Probability and Introduction to Statistics

Slides for Week 11-1

Topics: Moment Generating Functions

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Moment-Generating Functions

Moment-Generating Function is a very useful tool that can simplify certain calculations dramatically.

The **moment-generating function (mgf)** of a random variable X is

$$M(t) = \mathbb{E}(e^{tX}),$$

if the expectation is defined.

$$M(t) = \begin{cases} \sum e^{tx} p(x), & \text{discrete case} \\ \int_{\mathbb{R}} e^{tx} p(x) dx, & \text{continuous case.} \end{cases}$$

- $M(t) = \mathbb{E}(e^{tX})$ **may or may not exist** for a particular value of t .

Theorem

Suppose the moment generating function exists for t in a neighbourhood of 0 (an open interval containing 0). Then it *uniquely* determines the probability distribution.

Proof is beyond the scope of this course.

Differentiate $M(t)$ is

- X is discrete
- X is continuous

MGF of Binomial

Find the mgf of $X \sim \text{Bin}(n, p)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$\begin{aligned} M(t) &= \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^n \binom{n}{k} (pe^t)^k (1-p)^{n-k} \\ &= (pe^t + 1 - p)^n \end{aligned}$$

The sum converges for all t . By differentiating, we have

$$M'(t) = n(pe^t + 1 - p)^{n-1} pe^t$$

$$M''(t) = n(n-1)(pe^t + 1 - p)^{n-2} (pe^t)^2 + n(pe^t + 1 - p)^{n-1} pe^t.$$

Evaluating these derivatives at $t = 0$ gives

$$\mathbb{E}(X) = np, \quad \mathbb{E}(X^2) = n(n-1)p^2 + np,$$

and so $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = np(1-p)$.

MGF of Poisson

Find the mgf of $X \sim \text{Poisson}(\lambda)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$\begin{aligned} M(t) &= \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} e^{-\lambda} \\ &= e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}. \end{aligned}$$

The sum converges for all t . Differentiating, we have

$$\begin{aligned} M'(t) &= \lambda e^t e^{\lambda(e^t - 1)} \\ M''(t) &= \lambda e^t e^{\lambda(e^t - 1)} + \lambda^2 e^{2t} e^{\lambda(e^t - 1)}. \end{aligned}$$

Evaluating these derivatives at $t = 0$ gives

$$\mathbb{E}(X) = \lambda, \quad \mathbb{E}(X^2) = \lambda^2 + \lambda,$$

and so $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \lambda$.

When X is continuous

$$M'(t) = \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} p(x) dx = \int_{-\infty}^{\infty} x e^{tx} p(x) dx$$

and

$$M'(0) = \int_{-\infty}^{\infty} x p(x) dx = \mathbb{E}(X).$$

In general,

By differentiating $M(t)$ r times, we have

$$M^{(r)}(0) = \mathbb{E}(X^r).$$

Theorem

If the moment generating function exists in a neighbourhood of 0, then so do all the moments and $M^{(r)}(0) = \mathbb{E}(X^r)$.

MGF of Exponential

Find the mgf of $X \sim \text{Exp}(\lambda)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$M(t) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

By differentiating, we have

$$M'(t) = \frac{\lambda}{(\lambda - t)^2}, \quad M''(t) = \frac{2\lambda}{(\lambda - t)^3}.$$

Evaluating these derivatives at $t = 0$ gives

$$\mathbb{E}(X) = \frac{1}{\lambda}, \quad \mathbb{E}(X^2) = \frac{2}{\lambda^2}, \quad \text{Var}(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

MGF of Standard Normal

Find the mgf of $X \sim N(0, 1)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$M(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2x^2 + tx} dx.$$

The integral converges for all t and can be evaluated using the technique of completing the square.

$$-\frac{x^2}{2} + tx = -\frac{1}{2}(x - t)^2 + \frac{t^2}{2}.$$

Therefore,

$$M(t) = \frac{e^{t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx = e^{t^2/2} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx \right) = e^{t^2/2}.$$

We just obtained

$$M(t) = e^{t^2/2}.$$

By differentiating, we have

$$M'(t) = te^{t^2/2}, \quad M''(t) = e^{t^2/2} + t^2e^{t^2/2},$$

and

$$\mathbb{E}(X) = M'(0) = 0, \quad \mathbb{E}(X^2) = M''(0) = 1.$$

Hence

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = 1.$$